

Cultural Similarities and Differences in the Design of University Web sites

Ewa Callahan

School of Library and Information Science
Indiana University, Bloomington

This study examines cultural differences and similarities in design of university Web sites using Hofstede's model of cultural dimensions. Graphical elements on a sample of university home pages from Malaysia, Austria, the United States, Ecuador, Japan, Sweden, Greece and Denmark are compared using content analysis methods. The home pages were analyzed on the basis of two criteria: organization and graphical design. Element frequency scores were correlated with Hofstede's indexes and interpreted on the basis of the existing literature. The results suggest that similarities and differences in Web site design can be brought out through Hofstede's cultural model. Computed correlations between Hofstede's scores and frequency counts of interface elements were weaker than anticipated, but in most cases occurred in the hypothesized direction.

doi:10.1111/j.1468-2958.2006.00012.x

Introduction

The global ubiquity of information and communication technologies (ICTs), and the Internet in particular, raises the question of what influence those technologies will have on different regions and cultures. Are the technologies contributing to the homogenization of cultures around the Western model, or are other cultures using the technologies in culturally specific ways, making the Internet as varied as the real world?

This general concern gives rise, in turn, to a number of empirical research questions: How do different cultures represent themselves on the Web? Are the designs of web sites in different cultures distinct or do they follow similar trends? If cultural differences are found, what role do they play, and what are their practical consequences? This study attempts to shed light on how the designs of web sites vary among cultures, through the use of the cultural model of Dutch anthropologist Geert Hofstede.

Cultural differences in design have been studied at length in transnational marketing research (e.g., Albers-Miller & Gelb, 1996; Cutler & Javalgi, 1992), in order to discover the most effective advertising methods and strategies to target people in other countries. These studies have found significant differences in how products are

presented to customers. Cutler and Javalgi (1992) found differences in advertisement size, use of black and white visuals, frequency of use of photographs, and use of metaphor between U.S. and European countries. Al-Olayan and Karande (2000) discovered differences in the frequency of figurative images, comparative images, and appeal (emotional vs. rational) between Arab countries and the United States. Preferences for different advertisement themes were discovered by Tansey and Hyman (1990) in Brazilian and U.S. car advertisements (cf. similar research collected by Hermeking, this issue).

Advertising strategies have been utilized in university contexts for a long time, e.g., in the form of posters, flyers, and promotional materials to create and emphasize the image of the university. The university becomes a commodity and its prospective students become consumers (Wernick, 1991). Using promotional materials, universities attempt to describe the benefits that come through association with their institution. As an example, Wernick (1991) describes a poster for Trent University and explains the symbolism of its iconography which highlights Tradition as equated with Quality, a professional but informal faculty, solidarity, and the beauty of the environment, which together present the university as a nurturing and challenging environment. The promotional iconography, however, is not homogeneous throughout the entire university domain; it varies depending on the commodities the universities want to promote. While one institution may promote its academic achievements, another may advertise its exceptional sports record to present itself as an institution that encourages not only learning but overall development.

The Web creates a unique opportunity in that promotional materials can be accessed instantaneously from all over the world. Many companies and organizations consider their home page a virtual gateway to the world. Similarly to product advertising, organizations are using visual techniques to present their image to potential customers. If the customer base is international, knowing and acknowledging the aesthetic preferences of transnational clients on organizational web sites shows respect for the clients' cultures and might increase a company's contacts, opportunities, and profits. Similarly, universities use the Web as an advertising forum to appeal to local and international students.

The international student market varies for each country; thus it can be assumed that there is variation in the emphasis universities place on creating foreign language versions of their Web pages. Study in a foreign country requires working knowledge of the language of instruction. Since English is a commonly spoken first language (and the most commonly spoken second language in the world), American, British, and Australian universities might reasonably be expected to have greater numbers of potential foreign students. Universities in countries speaking other languages with a large number of speakers worldwide, such as Spanish and Arabic, would attract students within their linguistic domains. The need for international visibility is not limited to the competition for students, but also includes creating an image of the university in the international arena. Even universities in countries which, because of the uniqueness of their language, will not attract many foreign students, may publish

Web pages in English to facilitate academic exchange of ideas and create greater interest in the academic environment.

With increased competition, Web sites designed according to principles of effectiveness and efficiency of interaction and customer satisfaction may attract more viewers. The large international market raises the question as to whether the Web pages designed in one country are equally appealing to potential consumers in other countries. Cultural differences have become an important issue in international interface design, yet most publications on this subject concentrate narrowly on guidelines for the internationalization of the interface according to the country's and/or region's standards for language, date, measurements, currency, spelling, etc. (del Galdo, 1990; Russo & Boor, 1993). Less attention is placed on cultural differences in Web site organization and visual representation. Studies show, however, that information organization varies among countries. For example, Schmid-Isler (2000) examined news sites and noted the difference in site organization between Chinese- and Western-style home pages: Chinese home pages were often divided into many independent spaces, while on Western-style Web sites the layout was arranged around a focal point of the page. These layouts are based on culturally-dependent schemata of information storage and display (*browse vs. focus*). Similarly, Dormann and Chisalita (2002) distinguished between "visual" (based on graphics) and "index" (link-based) sites. (See also Bucher, 2002.)

Another aspect of Web design—color preferences in different cultures—is also widely discussed in the HCI literature, mostly from the perspective of color symbolism (del Galdo, 1990; Russo & Boor, 1993). Cultural color preferences were also observed in empirical studies. Barber and Badre (1998) noticed that the colors of the national flags were usually used by governmental sites in all the countries they studied, with the exception of Brazil, which used a variety of bright colors. French sites heavily used blue, white, and red (the colors of the French flag) regardless of the domain, while Israeli and Lebanese Web sites frequently included the color green.

Duncker, Theng, and Mohd-Nasir (2000) compared differences in color choices among Web sites created by students with various ethnic backgrounds. English students tended to use pastel colors, and frequently utilized gray, while Scandinavians favored dark hues. Students with a Jamaican background chose bright colors and used high contrasts. Black as a background color and colorful elements were frequently chosen by African students, while European and U.S. students tended toward bright backgrounds.

Cultural differences in image modality were also reported by Rajkumar (2003), who compared university Web sites in India and the United States. Images on United States Web sites tended to be high quality, with high saturation of color, while Indian images were washed out, had duotone effects, and low color saturation. Lin (2003) found differences in color contrast between U.K. and U.S. Internet bank Web sites; U.K. banks used stronger contrast of colors than U.S. sites. She also noted differences in the use of color schemes throughout the sites: Japanese Internet banks in Lin's study used the same base colors for every page except the home page, while U.S. and

U.K. sites tended to have the same colors throughout the whole site, or used different color schemes for different groups of options (color coding).

Information presentation (layout, graphics, color, etc.) is important, since it influences Web site usability. Tractinsky (1997) found correlations between Japanese and Israeli subjects who judged aesthetics and perceived usability of Japanese and Israeli interfaces. Shepard and Scholtz (1999) noticed that design of the sites using cultural markers increased performance by members of that culture. In studies by Choong and Salvendy (1999), the thematic structure of information and concrete knowledge representation reduced performance time as well as the error rate for Chinese participants, and a functional interface reduced the number of errors in the American group. (See also Faiola & Matei, this issue.)

Hofstede's Model of Culture

Several studies (Dormann & Chisalita, 2002; Marcus & Gould, 2000; Robbins & Stylianou, 2002) have tried to explain cultural differences in Web site design through the works of Geert Hofstede (Hofstede, 1980, 1991, 2001, 2002), who argued that world cultures vary along five consistent dimensions: power distance (PD), individualism vs. collectivism (IC), femininity vs. masculinity (MAS), uncertainty avoidance (UA), and long- vs. short-term orientation. Hofstede's dimensions may be characterized as follows:

Power distance refers to the degree of acceptance of the unequal distribution of power in society. High PD countries like Malaysia, Guatemala, and Mexico are characterized by tall hierarchies in organizations, in which the relationships between superiors and subordinates are stricter than in low PD countries (Austria, Israel, New Zealand). In low PD countries, subordinates are more likely to express their opinions and participate in managerial decisions. This distinction carries over to other levels of society. For example, in high PD countries, older generations are revered and parents and teachers demand obedience and expect respect from children, while in low PD countries children are treated as equals and are taught to be independent from an early age.

Individualistic societies (the United States, Australia) value personal achievement while collectivistic ones (Panama, Ecuador) emphasize the benefits of working in a social group. People in individualistic countries tend to take care of themselves and their nuclear family. In *collectivistic* cultures people are integrated in strong groups, which offer protection in exchange for loyalty. The meaning of the family is extended to all blood and in-law relatives. The accomplishments of the group are valued—a person should use his or her skills to advance the group.

Cultures with a high *masculine* index maintain traditional distinctions between gender roles and perceptions, whereas *feminine* cultures tend to dissolve gender differences. Masculine cultures, such as Japan and Italy, place values on challenge,

advancement, social recognition, and acquisition of wealth, while feminine cultures (e.g., Norway, Sweden) value quality of life, security, taking care of others, and the environment.

Uncertainty avoidance measures the degree to which people tend to stay away from uncertain situations. It is often mistakenly interpreted as risk avoidance, but actually avoiding uncertain situations could be accomplished by setting strict rules, or by actually taking risks to prevent long term uncertainty. High uncertainty avoidance countries (e.g., Greece, Portugal) prefer formal rules and regulations in order to reduce the amount of uncertainty, and are less inclined toward change than countries with a low uncertainty avoidance index (e.g., Jamaica, Singapore).

Hofstede defines the fifth dimension, *long/short-term orientation*, as follows:

Long Term Orientation stands for the fostering of virtues oriented towards future rewards, in particular, perseverance and thrift. Its opposite pole, Short Term Orientation, stands for the fostering of virtues related to the past and present, in particular, respect for tradition, preservation of “face” and fulfilling social obligations. (Hofstede, 2001 p. 359)

These dimensions are not distinct; they overlap and correlate with each other to a certain degree.

Hofstede’s research was conducted in international subsidiaries of IBM between 1967 and 1973, and was based on a survey, the results of which were entered into an IBM database. The primary goal of the survey was to research work attitudes of international employees. The questions asked can be divided into four categories: 1) satisfaction, related to personal evaluation of the work situation; 2) perception, related to subjective descriptions of an aspect or problem of the work situation; 3) personal goals and beliefs, related to a subject’s ideal job or to general issues in the industry, and 4) demographics such as age, gender, years of education, and years with the company. Most of the questions used 5-point scales. Fifty countries and three regions were surveyed, some of them by Hofstede himself.

Although the IBM survey was conducted in organizational contexts, Hofstede’s model of culture is often used in cross-cultural research to interpret a large variety of research findings. According to the Social Science Citation index, Hofstede’s books *Culture’s Consequences* (1980) and *Cultures and Organizations* (1991) have together been cited over 3,500 times since their first publication and are used in a large variety of disciplines: cross-cultural and organizational psychology, sociology, management, and communication (Callahan, 2004).

Hofstede’s theory of cultural dimensions has sparked several criticisms over the years. Critics have questioned the methodology used and the validity of the data. In his study of reviews, citations, and replication of Hofstede’s research, Søndergaard (1994) identified a number of articles which pointed to three major constraints in Hofstede’s study: Some argued that data collected between 1968 and 1973 are no

longer valid; others questioned the use of employees of one company as a basis for conclusions about national dimensions; yet others contended that the use of attitude surveys does not constitute a valid basis for this type of research. Others, like Baskerville (2003), question the use of nations as a unit of cultural analysis. Fang (2003) reviewed and criticized the merits of Hofstede's fifth dimension.

The most interesting debate, sparked by an essay by Brendan McSweeney (2002), has taken place in the pages of the journal *Human Relations*. McSweeney undermined Hofstede's dimensions by arguing that they are based on flawed assumptions. Specifically, McSweeney criticized the assumptions that (1) IBM organizational culture is homogenous across nations, so that the observed differences in culture can be taken as exclusively the results of upbringing in national cultures; (2) employees of one company are representative of the whole nation; (3) main dimensions of national culture can be identified by a questionnaire; (4) culture can be identified by response difference analysis; and (5) people's behavior in the workplace is mirrored in other areas of their lives. McSweeney's criticisms were answered by several researchers, including Hofstede himself.

McSweeney's argument that Hofstede's dimensions are not situationally specific was addressed by Smith (2002) with regard to the individualism-collectivism dimension. Smith argued that Hofstede's research shows that some work goals require more individualistic approaches than others, but this does not imply that the work goals could be equated with other life goals. Smith also did not agree with Hofstede's argument that power distance indices are very closely correlated to those of individualism-collectivism and that the major difference which separates those dimensions is Gross Domestic Product (GDP). Smith argued that there is no sufficient data to support this claim, especially since there is no longitudinal research on this issue. Smith presented his view along with other critical voices of Hofstede's study, especially Roberts and Boyacigiller (1984), who argued the measurement of validity was the major problem in Hofstede's research.

In this debate, Williamson (2002) adopts a mediating standpoint, arguing that Hofstede's perspective and McSweeney's view of national culture come from two different paradigms (interpretive vs. functionalist, respectively), and that both should be considered in social science research. Although he emphasizes the peril of assuming that all members of the culture are alike, and that all peoples' values and undertakings are culturally related (leading to problematic attempts to predict peoples' behavior and preferences based on the culture they come from), Williamson argues that disregarding Hofstede's approach would be a mistake since it provides a valuable framework for social scientists working outside the functionalist paradigm.

Hofstede himself replied to McSweeney's criticisms in the subsequent issue of *Human Relations* (Hofstede, 2002), pointing out that much of the criticism had already been addressed in the new edition of *Culture's Consequences* (2001). He acknowledged that surveys are not an optimal way of researching cultures, but since they are a common method in sociological research, they should not be disregarded as long as they are supported by other methods (Hofstede, 2001). Furthermore,

Hofstede argued that cultures do not change rapidly, so that even twenty-year-old data provide accurate results, as has been confirmed by numerous replication studies. He acknowledged that nations are not the best units for studying cultures; but because the boundaries of cultures cannot easily be determined, researchers do not have much other choice. In response to the criticism that a study of the subsidiaries of one company cannot provide information about entire national cultures, and its results cannot be generalized beyond the scope of the organizational subculture, Hofstede responded as follows: "What were measured were *differences between* national cultures. Any set of functionally equivalent samples from national populations can supply information about such differences. The IBM set consisted of unusually well matched samples of an unusually large number of countries" (Hofstede, 2001, p.73). Hofstede also remonstrated with McSweeney for not reading his work carefully enough, since Hofstede's 1980 book contains over 90 independent significant correlations which validate the study. In the 2001 edition, the number of independent significant correlations was increased to 400, based on analyses by other researchers who have used Hofstede's model, including several large-scale replication studies, which thus provide additional validation of his work. This argument is, however, weakened by the fact that in his book Hofstede mostly calls on studies that support his theory, without devoting much attention to studies that do not confirm his findings.

Hofstede's model has also been used by Human-Computer Interaction (HCI) researchers to explore similarities and differences in the design of Web sites in different cultures. Marcus and Gould (2000) discuss how Hofstede's dimensions can be related to the graphical organization and linguistic aspects of a site. They suggest that Web sites in high PD countries will have highly structured access to information, tall hierarchies of mental models, and a significant emphasis on social and national order in their symbols, on security and restrictions of access, and on the prominence given to leaders. In contrast, less structured access to information, shallow hierarchies, fewer access barriers, and a lesser focus on expertise, authority, logos, official stamps, etc. will characterize Web sites in countries with low PD. Frequent images of success, emphasis on change and progress, and the presence of personal information will be characteristic of highly individualistic countries. Websites in collectivist countries will present sociopolitical achievements and emphasize experience, history, and tradition. Masculine interfaces will focus on tasks and the efficiency of their completion, navigation will be oriented toward exploration and control, and interactive elements like games and animations will also be frequent. Feminine interfaces will support cooperation and exchange of information: The emphasis will be placed on visual aesthetics.

With regard to the uncertainty avoidance dimension, interfaces in countries with a high UA index will be simple with clear metaphors and limited choices; low UA Web sites will be more complex. With regard to time, short-term oriented countries will emphasize design that will allow user to accomplish tasks quickly. [Readers should

compare these general predictions with the empirical findings presented in this issue by Elizabeth Würtz—Ed.]

Another set of Web characteristics for each of the Hofstede's dimensions was identified by Robbins and Stylianou (2002), who studied commercial Web sites in several geographical regions. They conducted frequency counts of web site elements. According to the researchers, PD would be indicated by the presence of organizational charts, bio-sketches of top leaders, and messages from CEOs. Uncertainty avoidance would be represented by listings of job openings, cookie disclosers, etc. Site registration requirements, security provisions, and privacy policy statements would denote individualism/collectivism and annual reports and financial highlights represent masculinity/femininity on the site. Long/short-term orientation would be indicated by the presence or lack of a search engine, site map, FAQ, corporate history, etc.

Another study that used Hofstede's dimensions to evaluate Web sites was conducted by Dormann and Chisalita (2002). They examined university Web sites in five counties on the masculinity/femininity index. The authors did not conduct the evaluation themselves but rather designed a questionnaire and asked participants from the Netherlands (feminine culture, low MAS index) and Austria (masculine, high MAS index) to evaluate Web sites from countries with different values on the Masculinity index. Study participants rated higher all the feminine values in feminine Web sites, but with regards to masculine values only authority and toughness were rated higher in home pages from masculine countries. The study found a limited correlation between the MAS index and masculine and feminine values presented on the sites. The study is additionally important in that it examines the issue from the perspective of people of different cultures rather than from Web site elements as such.

Rajkumar (2003) examined Indian and U.S. university Web sites in relation to uncertainty avoidance, individualism/collectivism, and long-term orientation. According to Hofstede's indices, the two countries are similar in terms of uncertainty avoidance (India: 40, U.S.: 46), but differ along the individualism/collectivism dimension: The U.S. scores high on individualism, while India shows more collectivism tendencies (U.S.: 91, India: 48). India also has greater long-term orientation (61) than the U.S. (29).

The results of Rajkumar's study suggest that the differences in those three dimensions are visible in the Web site design. U.S. university Web sites showed a high degree of individualism throughout by way of frequent pictures of individuals, direct address (using "you" as opposed to "we"), personalization features, expression of private opinion, individual success stories, etc. Indian sites displayed images of groups, used formal speech, included mission statements that impacted the larger group, and stated opinions on group behavior. United States sites had few references to tradition—consistent with the low score in long-term orientation—emphasized current news and events, and presented clear strategic plans and step-by-step guides. Indian sites emphasized tradition and history through symbolic images, provided

archives of early convocational photos and images of founders, and made frequent references to the distant future. [These findings should be compared with E. Würtz's analysis of Indian Web sites, this issue. With regard to the use of "I" vs. "we" in individualist vs. collectivist cultures, see Leonardi, 2002. With regard to more formal language vs. informal language, cf. Evans, McBride, M., Queen, A., Thayer, & Spyridakis, 2004—Ed.]

Interestingly, in contrast to Hofstede's index values, the results for uncertainty avoidance show that Indian sites provide more evidence for the dimension, including formal organization charts, listings of rules and regulations, extensive legalese and other guidelines. By contrast, only 20% of U.S. sites included such information—usually related to SARS and emergency procedures.

The existing literature suggests that Hofstede's dimensions of culture are indeed visible in interface design. However, in most of the studies the authors assigned the indicators for each dimension hypothetically, based on similar values representative of a specific dimension. As a consequence, there is no verification that the assigned site characteristics are really indicators of the dimension. This study investigates whether the graphical elements mentioned in the literature are in fact the best markers for a specific dimension, by examining the correlations between graphical elements and Hofstede's index values for selected countries. The Web sites analyzed show correlations with Hofstede's dimensions, although these are much weaker than initially supposed. In addition to characteristics of national culture, other factors—genre, available technology, institutional guidelines, etc.—are found to affect Web-site design.

Research Question and Hypotheses

The question that guides this study is to what extent the design of Web sites differs across cultures, and whether the differences can be measured using Hofstede's cultural model. To investigate this question, the genre of university Web sites has been chosen. A single genre was selected since Web sites from different genres usually contain different content and visual elements (Barber & Badre, 1998); conducting the study on a variety of web genres would introduce too many variables to arrive at meaningful results. I further narrowed the scope of the study by limiting the observation to graphical elements only, since conducting linguistic analysis would require proficiency in the languages of the Web sites.

The genre of university Web sites was chosen for several reasons. As Rajkumar (2003) points out, education is socially and culturally shaped. Thus, cultural values and preferences should be reflected in the goals and objectives of universities and echoed in the design of their Web pages. At the same time, universities cater to the young in their countries. By virtue of their educational mission, they bestow cultural values. The third reason is that universities in today's world, as institutions of education and culture, seem to be well aware of the opportunities the Internet brings for the development of wider international relations. To the extent that this is true,

the design of the home page as a virtual gateway to the rest of the world should be given high priority.

To address the research questions stated above, the available literature on the use of Hofstede's cultural model in the interpretation of differences in graphical design was first examined. Table 1 below summarizes this literature. Since the short/long-term dimension was formulated by Hofstede on the basis of a Chinese Cultural Survey much later than the first four dimensions and was tested on another set of countries, it was excluded from the present analysis.

Table 1 Web site characteristics in relation to Hofstede's dimensions of culture

Dimension	High	Low	Source
Power distance	• symmetry • tall hierarchies • focus on official seal, national emblems • photographs of leaders • monumental buildings • monuments	• asymmetry • shallow hierarchies • photos of students rather than faculty • images of both genders • images of public spaces • images of everyday activities	Marcus and Gould (2000) Ackerman (2002)
Individualism	• images of individuals • images of young • emphasis on action	• images of groups • images of aged and experienced • emphasis on state of being	Marcus and Gould (2000)
Masculinity	• limited choices • orientation toward goals • emphasis on tradition and authority • frequent images of buildings • graphics used for utilitarian purposes	• multiple choices • orientation toward relationships • frequent images of people, especially showing them laughing, talking or studying together • attention gained by visual aesthetics	Ackerman (2002) Dormann and Chisalita (2002) Marcus and Gould (2000)
Uncertainty Avoidance	• limited choices • restricted amounts of data • limited scrolling • references to daily life • redundancy	• variety of choices • long pages with scrolling • abstract images	Marcus and Gould (2000) Ackerman (2002)

Based on these observations, the following hypotheses are advanced:

- H1: The frequency of symmetrically designed sites is correlated positively with country scores on the PD dimension. Thus, countries with a higher PD index will have a symmetrical home page more frequently than countries with a lower PD index.
- H2: The relative frequency of logos is correlated positively with country scores on the PD dimension. Thus, home pages in higher PD countries will display logos of the university more frequently than home pages in lower PD countries.
- H3: The relative frequency of images of faculty is correlated positively with country scores on the PD dimension. The relative frequency of images of students is correlated negatively with country scores on the PD dimension. Thus, countries with a higher PD index will have images of the faculty more frequently than lower PD countries; students will be portrayed more often in lower PD countries than in higher PD countries.
- H4: The relative frequency of images of buildings is correlated positively with country scores on the PD dimension. The relative frequency of images of people is correlated negatively with country scores on the PD dimension. Thus, high PD countries will depict buildings more frequently than lower PD countries, which will favor displaying images of people.
- H5: The relative frequency of images of individuals is correlated positively with country scores on the individualism/collectivism (IC) dimension. The relative frequency of images of groups is negatively correlated with country score on the individualism/collectivism dimension. Thus, individualistic countries (high IC) will more likely present pictures of single people, while collectivist countries (low IC) will more likely present people in groups.
- H6: The relative frequency of images of young individuals is correlated positively with country scores on the IC dimension. The relative frequency of images of older individuals is correlated negatively with country scores on the IC dimension. Thus, countries with a higher IC index will have a higher frequency of pictures of students, while countries with a lower IC will have a higher frequency of pictures of faculty.
- H7: Number of links is correlated negatively with country scores on the masculinity/femininity dimension. Thus, countries with a higher masculinity index (masculine countries) will have fewer links per page than countries with a lower masculinity index (feminine countries).
- H8: The relative frequency of figurative images is negatively correlated with country scores on the masculinity/femininity dimension. The relative frequency of images of buildings is positively correlated with country scores on the masculinity/femininity dimension. Thus, feminine countries will more likely present figurative images, while masculine countries will more frequently present buildings.
- H9: The relative frequency of high modality pictures is correlated positively with country scores on the masculinity/femininity dimension. The relative frequency of low modality

pictures is correlated negatively with country scores on the masculinity/femininity dimension. Thus, masculine countries will have a higher frequency of high modality pictures than feminine countries, while feminine countries will have a higher frequency of low modality pictures than masculine countries.

H10: The relative number of pictures of women is correlated negatively with country scores on the masculinity/femininity dimension. Thus, masculine countries will have fewer pictures of women than feminine countries.

H11: The relative frequency of animated pictures is correlated positively with country scores on the masculinity/femininity dimension. Thus, masculine countries will have more animated pictures than feminine countries.

H12: Number of links is correlated negatively with country scores on the UA dimension. Thus, countries with a higher UA score will have fewer links than countries with a lower UA score.

H13: The frequency of horizontal pages is correlated positively with country scores on the UA dimension. Thus, horizontal pages will be more frequent in the countries with high UA than in countries with low UA.

H14: The relative frequency of abstract images is correlated negatively with country scores on the UA dimension. Thus, abstract images will be more frequent in the countries with lower UA than in countries with higher UA.

The notion of relative frequency of different types of images refers to the proportion of this type of image in relation to the total number of images on the Web sites examined in the particular country. The exception is logos, where relative frequency relates to the number of logos in relation to the number of sites in a country.

Study Description

Country Selection

For each of the dimensions of culture identified by Hofstede, two countries were selected, one with the highest score and one with the lowest score in Hofstede's research. In two cases the number of universities in the selected countries was not large enough to allow analysis. Guatemala, the country with the lowest IC index, has only seven universities; Singapore, lowest in UA, has only three universities. In those two cases Ecuador and Denmark, the first countries on the lowest ends of the dimensions that satisfied the selection criteria, were chosen instead.

Web site Selection

Only institutions which offered bachelors and/or masters degrees (or equivalent) were selected, regardless of the title of the institution itself, in order to preserve the integrity of the sample. To arrive at a comprehensive list of the universities in each of the chosen countries, the following procedure was applied. First, the Web site of the

Table 2 Numbers of university Web sites in selected countries

Dimension	Index Value	
	High	Low
<i>Power Distance (PD)</i>	Malaysia (28)	Austria (74)
<i>Individualism/Collectivism (IC)</i>	USA (1547)	Ecuador (55)
<i>Masculinity/Femininity (MAS)</i>	Japan (101)	Sweden (41)
<i>Uncertainty Avoidance(UA)</i>	Greece (21)	Denmark (248)

country's higher education governing body was searched for the complete list of universities. This gave positive results for Austria (Federal Ministry for Education, Science and Culture), Japan (Ministry of Education, Culture, Sports, Science, and Technology) and Ecuador (National Council for Higher Education). In the case of Sweden, the United States, Denmark, and Greece, ministerial sites did not have lists of the universities, but offered links to some external sites that provided this information. The lists of the universities were entered into a Microsoft Excel spreadsheet, and each university was assigned a number. The RANDBETWEEN function was used to randomly select 20 universities from each country.

Analytical Methods

Content analysis methods were applied to the home pages of each of the universities, with an emphasis on methods of visual content analysis (Bell, 2001). Because I was interested in finding cultural markers typical of specific cultures, only the pages in the cultures' native languages were examined (with the exception of when the website was in English only), since in some cases the English-language pages were designed differently (and underdeveloped) compared to their native language counterparts.

The home pages were analyzed on the basis of two criteria: organization and graphical design. First, several pages were analyzed to establish a preliminary set of categories, which were then applied to a small sample of university home pages. This allowed the discovery of categories missed in the first sweep. Since the number of different types of graphical elements is rather large, not all possible elements have been categorized. The categories were chosen according to the criterion of whether they could be examined empirically; also, only the elements that in the existing literature were identified as possibly being an indicator of one of Hofstede's dimensions were taken into consideration. A language category was added since language is an aspect of culture; this category was expected to show how much importance the universities place on reaching an international audience. The category "photo" was added on the basis of a study by Cutler and Javalgi (1992) that discovered significant differences between the presence of drawings and photos in U.S. and European print advertising. Table 3 lists the categories used in the study.

After the categories were identified, 10% of the sites (two sites from each country) were randomly selected again and coded by two separate coders to ensure reliability of the results. Overall initial agreement was reached at the 85.1% level,

Table 3 Category descriptions: Web page

Category	Description	Possible coding
Entry page	A page that does not contain any links except of the link to the home page. Also pages that contain only links to multiple linguistic versions of the home page.	Yes/No
Page orientation	The Web page design is organized to fit the screen (horizontal), or allows vertical scrolling. (Since the screen sizes vary, the page orientation was determined on the basis of the intention of the designer to fit the information on one screen, rather than physical properties of the site.)	Vertical/Horizontal
Symmetry	The Web page design is symmetrical or not.	Yes/No
Page organization	Description of how the information on the page is organized.	Description Ex. The page is divided into four sections: one horizontal at the top (contains the name of the university and the logo), next two columns (one with a menu) followed by the fourth, horizontal section
Menus	Description of the menus on the page.	Simple—one level menus: all options visible on the screen Complex—menus constructed of several levels: after user places a pointer on the option more choices appear
Number of links	Total number of links (including those on the menus) are counted for each page.	Numerical
Number of pictures	Total number of pictures (photos and graphics) on the homepage. Buttons or menus do not count as graphics, but photos used as links do.	Numerical
Color scheme	The color of the background and the most prominent colors on the page are noted.	Description

Table 4 Category description: image

Category	Description	Possible coding
Collage	A photo composed of photos or graphics in which separate border for each of the components cannot be distinguished. Pictures composed of several photos or graphics when the borders of each photo can easily be seen would be counted as single pictures.	Yes/No
Animated	All animated pictures as well as slideshows activated on download or mouse click.	Yes/No
Modality (Bell, 2001)	Degrees of modality: <i>High</i> —highly saturated colors, clear pictures	High
	<i>Medium</i> —medium and low saturated colors, blurred pictures	Medium
	<i>Low</i> —black and white or two tonal images	Low

with the major discrepancies being in the *modality* and *type* categories, where modern-style logos were counted by a second coder as abstract due to their contrasting appearance with seal-type logos. The categories of *modality* and *logo* were redefined accordingly, and the entire sample was coded.

Each picture was analyzed and assigned to one of the following categories: logo, person, building, art, nature, or artifact. If the picture depicted a human it was also categorized as “figurative.” This category included persons as well as logos and art pieces that depicted a human being. Figurative images were analyzed further according to the number (single, group), gender (male, female, mixed, unidentified), status (student, faculty, mixed, unidentified), activity (descriptive) and social distance (intimate, close personal, far personal, close social, far social, public) of the figures. Social distance as described by Bell (2001) relates to how a person is portrayed in photographs and drawings. *Intimate* depiction shows the head only, *close personal* shows the head and shoulders, *far personal* shows the person above the waist, *close social* shows a whole person, and *far social* shows a whole person with a background. *Public* refers to a group of people shown from the waist up.

After the coding was completed, the frequencies of occurrence of each category were counted and compared across the countries. For each dimension, the index calculated by Hofstede was recorded, and the correlations between each dimension and each category were calculated. Table 5 summarizes the indices and rankings of the selected countries in Hofstede’s studies.

Findings

Language

Observations regarding efforts of the universities to present themselves to international colleges and international students were included in the study to get a glimpse of how those institutions market themselves in the international arena. For this purpose, the language of the home page of each university was recorded as well as the presence of other linguistic versions of the home pages. In the case of bi/multilingual sites, the different linguistic versions were compared to determine if they were identical. English version home pages were then examined to see what kind of information was presented to international Web page visitors.

As expected, U.S. Web sites were in English only, although Hope International University (CA) also offered a Korean language page. Ecuadorian sites (all but one) were not translated into English or other language, presumably because their potential clientele is recruited from Spanish-speaking countries. In the other countries, the percentage of multilingual versions varies. In the case of Sweden, all sites had at least

Table 5 Hofstede's ranks and index numbers for selected countries

Power Distance			Individualism/ Collectivism			Masculinity/ Femininity			Uncertainty Avoidance		
Country	Index	Rank	Country	Index	Rank	Country	Index	Rank	Country	Index	Rank
Malaysia	104	1	USA	91	1	Japan	95	1	Greece	112	1
Ecuador	78	8.9	Denmark	74	9	Austria	79	2	Japan	92	7
Greece	60	27.28	Sweden	71	10.11	Ecuador	63	13.14	Austria	70	24.25
Japan	54	33	Austria	55	18	USA	62	15	Ecuador	67	28
U.S.A.	40	40	Japan	46	22.23	Greece	57	18.19	USA	46	43
Sweden	31	47.48	Greece	35	30	Malaysia	50	25.26	Malaysia	36	46
Denmark	18	51	Malaysia	26	36	Denmark	16	50	Sweden	29	49.50
Austria	11	53	Ecuador	8	52	Sweden	5	53	Denmark	23	51

Table 6 Multilingual web pages in selected countries

Country	Multilingual pages		Identical	
	Number	Percent	Number	Percent
Austria	12	60	6	50
Denmark	19	95	12	63
Ecuador	1	5	0	0
Greece	17	85	14	82
Japan	19	95	6	67
Malaysia	6	30	4	67
Sweden	20	100	3	15
U.S.A.	1	5	0	0
Total	95	59	45	47

one additional linguistic version; similarly, 19 sites in Denmark and the same number in Japan had versions in other languages. Austria had 17 multilingual sites (plus two where links to an English version were not working). Greece followed with 12 multilingual sites. Interestingly, the Technological Educational Institute of Pireus titled the link to the English pages the “E.U. version,” emphasized by the European Union flag.

Most of the sites with another language version limited themselves to two languages (usually the native language and English), but in several cases the number of languages was greater. Aristotle University of Thessaloniki has the identical home page in seven languages (Greek, English, Spanish, Italian, French, Russian, German). Similarly, Stockholm University included links to English, German, French, and Spanish versions. Kristianstad University College in Sweden offered a link to the page in Chinese, as did Kyoto University of Education (Japan), although in the latter case the link was not working. In several cases the English version of the page was the first one to be displayed, possibly because the site sent a cookie to check from where the download request was generated.

The case of Malaysia is interesting, in that it was difficult to determine which of the versions of the home page was the one in the major language. Three languages were used in Malaysian Web sites. Most of the sites were in Malay, six of which had an English version of the home page. Six sites were in English only. In the case of International Islamic University, the main home page was in English with an additional page in Arabic. University Sains Malaysia provided pages in Malay, English and Arabic.

The pages in other languages were usually linked to the native language home page. Less commonly, the user could chose the preferred version from the entry page. In three cases, the home page itself was multilingual. In most cases, pages in

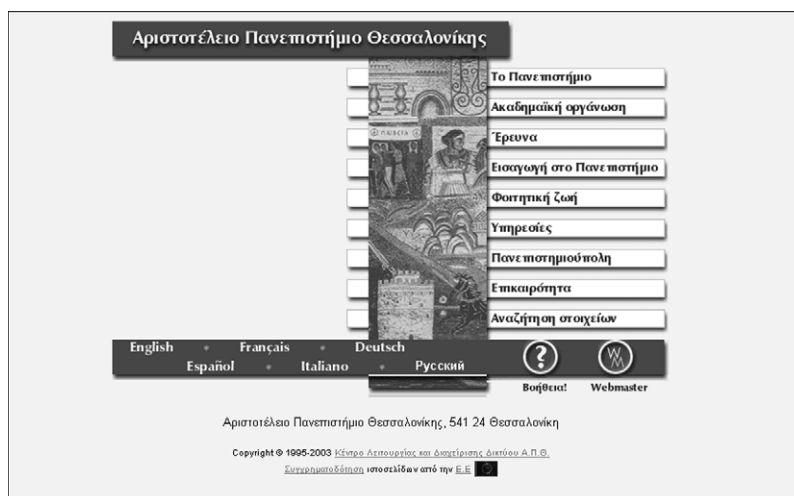


Figure 1 Home page of Aristotle University of Thessaloniki, Greece.

languages different from the native language consisted of a short description introducing the university and its mission to an international audience. Greece had the highest percentage of pages in identical linguistic versions, followed by Japan and Malaysia.

With regard to the content, English-language pages in Japan provided information similar to what one would expect to see on American Web sites: general information about the university, academics (often called *Organization*), admission, Information Technology, job offerings, etc., with the addition of links to International Exchange information, and overseas student admission.

English-language pages of Swedish universities were usually different from the official home pages of the university. They included some welcoming message or introduction to the university. Although top page menus generally had similar links as the main home page, the main menus had information pertaining to international students (International student admissions, courses and programs in English, European Credit Transfer, etc.) Lärarhögskolan i Stockholm (the Stockholm Institute of Education) even had links to visa and insurance information. Even though not all the universities in Austria provided English-language Web pages, the ones that did so offered information directly related to the needs of international students. Interestingly, the graphics on those pages were identical to the graphics on the German language home page; only the links were suitably tailored. Danish universities also followed this trend.

Greek international pages were, in the vast majority of cases, translations of the main home page, suggesting that there was no effort (or possibly no perceived need) to internationalize information about the university more broadly. The fact that the English-language pages are identical to the Greek ones raises the question of whether the information provided is that sought by foreign visitors, or whether it is a matter of convenience, in as much as translation is a relatively straightforward process compared to developing new content, which may involve multiple university units and require extensive coordination efforts. Addressing this question would require research on the universities' international markets, including interviews with Webmasters, possibly in conjunction with Web log analysis.

Color

In Greek university Web sites, two color schemes appear to dominate. Nine web sites were designed in various shades of blue, some of them relating to nautical scenes; 11 Web sites used various shades of brown and toned yellow—many of those sites used ancient Greek art as a central motif of the page. The colors on the Greek pages are rather toned down; only two sites displayed more vivid colors. White was most often chosen as a background color (13 times).

A white background was also commonly chosen for Malaysian sites (17); in three other cases the background was purple or blue. In contrast to Greece, vivid colors are dominant and the number of colors on the page is large. The Malaysian sites make heavy use of animation, mostly through animated text. Animation types vary from



Figure 2 Home page of Central University of Ecuador.

simple “new” buttons, through decorative flashing, scrolling news, and moving clickable menus.

Much like Malaysia, in Ecuador bright colors and animation play an important role. Some sites included many animated elements, and one had a large pulsing logo in the middle of the page. The use of white as a background color is much less common than in other countries; this color was only chosen in nine sites. Other choices included shades of blue, yellow, gray, and black.



Figure 3 Homepage of University College of Skövde, Sweden.



Figure 4 Home page of Saitama University, Japan.

Sweden displayed a strong preference for a white background (18 sites). The main characteristic of the Swedish sites was an absence of drawings. Virtually all the decorative elements that were not logos were photographs. Colors were used sparingly as a background for links. Blue and yellow, the Swedish national colors, are a frequent theme in addition to grey, purple, pink, and dark red.

In Denmark, color schemes involving various shades of blue seem to be dominant. Background choices were more diverse than in Sweden: ten white, six blue, three gray, one black. Most of the colors, except for rich blue, were toned down.



Figure 5 Johannes Kepler Universität Linz, Austria.

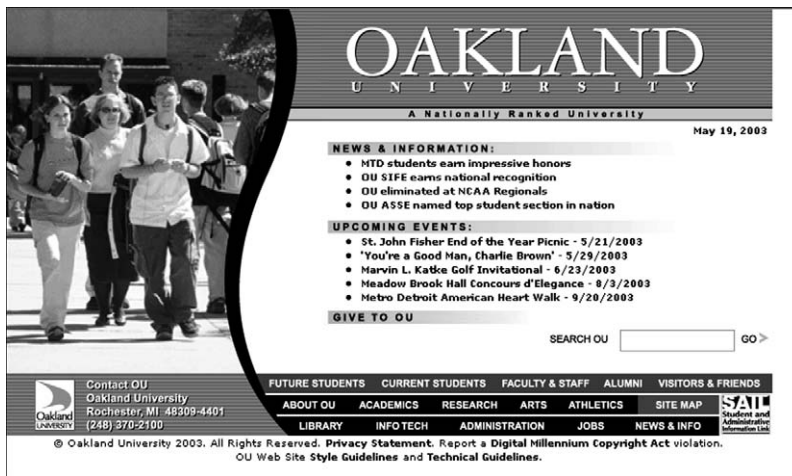


Figure 6 Home page of Oakland University, U.S.A.

In Japan, pastel colors dominated in the graphical elements and were also chosen for the background in the rare cases (4) when the background was not white.

The colors of the Austrian sites were mostly toned down. Four sites, however, had very bright colors. The background was white in 19 cases, and, on one bright site, rich blue. The color combinations varied from a conservative dark blue and white to rainbow multicolor. Similarly, in the U.S., the only two colors chosen for the background were white (17) and yellow (3). The colors for the graphic and textual elements, however, ranged from toned down to bright and from light to dark, with bold colors tending to be most prevalent.

Information Organization

Looking at the information organization data one can see that there are some culturally motivated differences in the ways universities in the selected countries organize their sites. Two of the most visible criteria are page orientation (horizontal vs. vertical), and the number of links presented on the pages. With regard to orientation, Austria, Denmark, and Ecuador clearly prefer horizontal page design, while Japan and Malaysia most frequently choose a vertical one. In the case of Japan it was difficult to determine the directionality of the pages since they often required both horizontal and vertical scrolling. The majority of the pages examined were asymmetrical, and none of the countries showed a strong preference for symmetry; on the contrary, Denmark and Sweden seem to avoid it.

Number of links varied significantly across the countries (ANOVA $F = 9.271$, $p = .000$).¹ The highest counts among the researched countries (Malaysia and Japan) were nearly twice as high as the numbers for Ecuador and Greece. A post hoc Bonferroni test revealed significant (at the .05 level) differences between Ecuador and all other countries but Greece. Significant differences (at the .05 level) were also observed between Japan and the countries of Austria and Greece.

The record number of links on a page (153!) belongs to Universiti Sains Malaysia. It is necessary to mention, however, the high variability in the number of links in the countries; with the exception of Greece, the Standard Deviation values are rather high.

Table 7 summarizes the characteristics of the organization of the home pages for each country. The numbers in columns “Horizontal page orientation” and “Symmetry” indicate how many out of the total number of home pages studied for each country (20) were horizontal (vs. vertical) and how many were symmetrical. For each country, the total number of links and pictures on all 20 pages was presented (Num.) together with the average number per page (M) and standard deviation (SD). The last column shows the total number of complex, pull-down menus on all the pages examined for that country.

The disparity between the highest and lowest numbers of links on a page in Malaysia is 145. However, there are differences in how the links were presented on the pages. While high counts of links in countries like Denmark or Sweden were the result of multiple layer menus, the most “linked” Malaysian site had all of its links in the body of the page.

Not surprisingly, the number of pictures on the page was also related to page directionality (Japan and Malaysia had the highest number of vertical pages and highest numbers of pictures per page), but the differences in the numbers of pictures between and within countries is much smaller than the differences in the number of links.

None of the countries showed preferences for complex menus. Most of them relied on links and simple menus, and in the case of Japan, almost entirely on links only. Most of the complex menus were limited to two levels; only in four cases (2 from Austria, 1 from Denmark, 1 from Greece) did the menus have three levels. The only four level menu in the sample was on the home page of Kolej Universiti Islam, Malaysia.

Image Analysis

The image analyses also led to interesting observations (see Table 8). Among the countries studied, Malaysia had an unexpectedly high number of collages, whereas

Table 7 Web page characteristics

Country	Horizontal page orientation	Symmetry	Links			Pictures			Complex menus
			Num.	M	SD	Num.	M	SD	
Austria	17	7	523	26.2	18.1	71	3.6	2.1	4
Denmark	18	3	628	31.4	27.5	73	3.7	2.8	3
Ecuador	17	8	476	23.8	19.4	77	3.9	2.8	2
Greece	15	5	452	22.6	10.0	57	2.9	2.7	1
Japan	4	6	891	44.6	20.1	80	4.0	3.4	0
Malaysia	8	6	990	49.5	39.7	90	4.5	2.6	2
Sweden	11	1	867	43.4	20.5	66	3.3	1.7	1
U.S.A.	13	0	781	39.1	15.6	68	3.4	2.1	1



Figure 7 The Danish University of Pharmaceutical Sciences.

Sweden and Denmark relied heavily on photos (it seems that logos were the only drawn graphical elements of the pages in Scandinavian countries), while Ecuador and Greece preferred drawn images. Animated images were popular in Ecuador and Malaysia. The high number of animated images in Austria is due to the fact that the site of Technische Universität Graz had several photographs which changed after

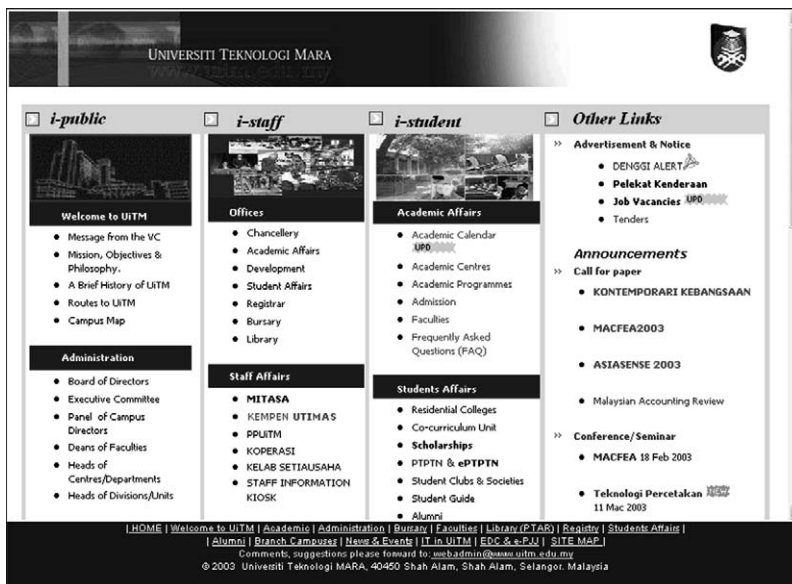


Figure 8 Technological University Mara, Malaysia.

several seconds. The animated images in the sites in Ecuador and Malaysia flashed at high speed. Some sites in those countries seemed to be pulsating since they also included many animated textual elements. U.S. sites often included a slide show, activated on reload or mouse click, so that the animations were not instantly apparent.

General image characteristics are summarized in Table 8. The numbers represent total occurrences on all the pages examined for the country; the percentages calculate the relation of the occurrences to the total number of pictures reported for the country.

Modality of the images was difficult to code, since the saturation of the colors can only be judged subjectively. With several colors in the pictures it was difficult to decide which one was dominant and should be coded. Additionally, the modality of the pictures was easier to determine in drawings than in photos. In drawings, the differences among the colors were more apparent because the designers can have better control over the contrast. In the case of photos, the colors of the picture depend on the colors of the photographed image and on the capability of the photographic software used.

In the case of photos, the modality of the picture depended more on the resolution. For example, Swedish photographs were high quality and were categorized as high modality, while on the Malaysian sites, the photos were not very sharp, so they were coded as medium modality, despite the bright colors. Of the countries studied, Austria, Greece, and the U.S. had the highest number of low modality images, while Malaysia, Sweden, and Ecuador had the highest number of high modality images.

The theme of the pictures also varied across countries. University logos were commonly present on the home pages, but only Malaysia had a tendency to include more than one per page. I observed that the older universities, regardless of country, used a university seal as a logo, while newer technical institutions designed more modern emblems. This observation was not coded in the present analysis, but would be interesting to explore in future studies.

Table 8 Image characteristics

Country	Total number of Images	Collage		Photo		Animation		Modality (%)		
		Number	%	Number	%	Number	%	High	Medium	Low
Austria	71	6	8.4	40	56.3	11	15.5	16.9	23.9	59.2
Denmark	73	1	1.4	54	73.9	4	5.5	23.3	47.9	28.8
Ecuador	77	5	6.5	25	32.5	11	14.3	35.1	53.2	11.7
Greece	57	2	3.5	22	38.5	0		10.5	43.9	45.6
Japan	80	1	1.2	38	47.5	6	7.5	7.5	60.0	32.5
Malaysia	90	22	24.4	50	55.5	9	10	43.3	51.1	5.6
Sweden	66	3	4.5	44	66.6	0		37.9	40.9	21.2
U.S.A.	68	0		43	63.2	8	11.7	17.6	33.8	48.5

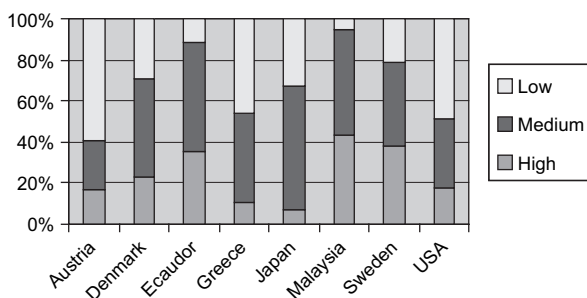


Figure 9 Image modality in chosen countries.

The most common themes in the images were people and buildings. People were the most popular subject in all countries except for Greece, where buildings and art were most prevalent. The most frequent theme of the Greek art was humans, however, which is reflected in the figurative category. Buildings were most common in Ecuador, Japan, and Greece, and artifacts in Greece and Austria. The most common artifacts were books, writing materials, computers, art (especially in Greece), flags, and maps.

Table 9 presents types of images found on the pages. The column *Figurative images* combines the number of images from the *Person* category and adds images from other categories that also depict humans. The percentages express the relation of the figurative images to the total images reported for the country.

Scandinavian countries had the highest percentage of figurative representations in their images, followed by Malaysia, Greece, and the United States. Of Malaysian figurative images, 83% showed groups of people, as did 74% of the images in Ecuador. The highest percentage of portraits of individuals is found for Greece (68%) followed by Japan (65%) and Sweden (61%).

The figurative images were analyzed further on the basis of gender, status, and social distance. Gender proportions were rather difficult to determine for several reasons. First, in Asian and South American cultures, the gender of the people in the pictures was more visible (hair length, clothing, head covering) than, for example, in Scandinavian countries. The large number of group pictures resulted in relatively high numbers in the *mixed* category, and pictures with far social or public distance also made judgments of gender more difficult. The collected data suggest that neither gender was especially favored overall. The higher number of females in Sweden is due to the fact that one of the sites presented multiple pictures of a design project in various stages, and the developer (female) was present in every picture. The high number of males in Greece relates again to Greek art rather than to representation of living people. The high number of mixed gender in Ecuador and Malaysia is due to the high number of images of groups in those countries.

Most of the sites, regardless of country, show students (again with the exception of Greece), although sometimes the status of the persons in the picture was difficult

Table 9 Image type

Country	Total number of images	Images by type						Figurative images	
		Logo	Person	Building	Artifact	Nature	Abstract	Number	%
Austria	71	15	19	12	13	2	10	19	26.8
Denmark	73	17	31	7	7	5	8	31	42.5
Ecuador	77	17	22	22	9	3	5	23	29.9
Greece	57	17	6	17	13	1	4	21	36.8
Japan	80	19	19	19	9	3	11	20	25.0
Malaysia	90	30	32	12	2	0	9	34	37.8
Sweden	66	17	30	7	8	2	2	33	50.0
U.S.A.	68	15	24	13	8	5	2	24	35.3

The sums of numbers in *Type* may be higher than the number of images since the depictions in collages could be included in several columns.

to determine. In only a handful of sites was there a picture which clearly indicated the status of persons as faculty (one in Malaysia, one in Japan, etc.). Faculty were more common in group pictures showing a class in session; however, those types of pictures were not very frequent.

The countries varied in the social distance the subjects of the pictures showed. Naturally the countries with a high number of group images will have a high number of public images (PUB), since those two categories are correlated by definition. Public images had the highest percentage in all countries except Japan, however, where close personal (CP) images were most popular, and the U.S., where 54% of the images showed far personal (FP) distance. Interestingly, Denmark and Sweden, countries that often are perceived as similar, since they both belong to Scandinavian culture, showed striking differences in this category. While Denmark's pictures (after public) were mostly far social (FS) and far personal distance, Sweden had a high percentage of intimate (INT) and close personal images.² The high number of

Table 10 Figurative images by gender

Country	Male		Female		Mixed		Unknown	
	Number	%	Number	%	Number	%	Number	%
Austria	4	21.1	1	5.3	2	10.5	12	63.2
Denmark	8	25.8	10	32.3	4	12.9	9	29.0
Ecuador	7	30.4	3	13.0	11	47.8	2	8.7
Greece	8	36.4	3	13.6	4	18.2	7	31.8
Japan	4	20.0	5	25.0	1	5.0	10	50.0
Malaysia	6	17.1	4	11.4	14	40.0	11	31.4
Sweden	4	12.1	12	36.4	9	27.3	8	24.2
U.S.A.	6	25.0	9	37.5	6	18.2	3	9.1
Total	47	22.7	47	22.7	51	24.6	62	29.9

Table 11 Figurative images by social distance

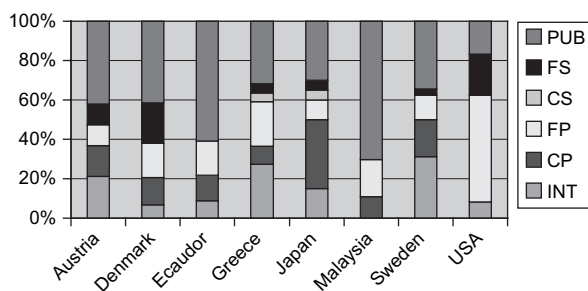
	Intimate		Close Personal		Far Personal		Close Social		Far Social		Public	
	Number	%	Number	%	Number	%	Number	%	Number	%	Number	%
Austria	4	21.1	3	15.8	2	10.5	0	0.0	2	10.5	8	42.1
Denmark	2	6.5	4	12.9	5	16.1	0	0.0	6	19.4	12	38.7
Ecuador	2	8.7	3	13.0	4	17.4	0	0.0	0	0.0	14	60.9
Greece	6	27.3	2	9.1	5	22.7	1	4.5	1	4.5	7	31.8
Japan	3	15.0	7	35.0	2	10.0	1	5.0	1	5.0	6	30.0
Malaysia	0	0.0	4	11.8	7	20.6	0	0.0	0	0.0	26	76.5
Sweden	10	30.3	6	18.2	4	12.1	0	0.0	1	3.0	11	33.3
USA	2	8.3	0	0.0	13	54.2	0	0.0	5	20.8	4	16.7

intimate pictures in Greece was due to the fact that most of its logos were figurative showing a profiled head; on other pictures far personal or public social distances were most common.

The number of intimate and close personal images in U.S. universities was rather small. Out of two intimate pictures, one was a poster of a missing student. Most of the images were far personal, followed by the far social category. These results contrast with the findings reported by Rajkumar (2003) for U.S. university sites. In his study, intimate and close personal images were most frequent. The discrepancy between those two studies may be due to the fact that Rajkumar's sample was smaller, including only ten sites.

Hofstede's Dimensions of Culture

The final step of the analysis correlated the frequency of selected categories with Hofstede's dimensions of culture. These correlations are presented in Table 12. Out of the 14 hypotheses articulated earlier in this article, only two were fully supported. The presence of logos was positively correlated (significant at the .05 level) with the PD dimension: The high PD countries had significantly higher frequency of logos than low PD countries. Logos, especially those based on the official seal of

**Figure 10** Social distance of figurative images by country.

university, are an institutional status symbol, so their correlation with the PD dimension is not surprising.

Images of people were negatively correlated with the masculinity/femininity index (significant at the .05 level), thus home pages in feminine countries (low MAS index) would more often show images of people than pages in high masculine countries. The significance increased when all figurative representations were considered.

Also, positive correlation of the MAS index with images of buildings was significant at the .05 level (countries with a high MAS index had more images of buildings), which is consistent with the observations of Dormann and Chisalita (2002). In Hofstede's terms, masculinity is associated with challenge and advancement, thus images of manmade monumental buildings may be common in high MAS countries, while pictures of people represent relationships associated with femininity.

Countries with high PD have great respect for authority, so images of faculty were expected to be significantly more common in those countries. The results show that images of faculty are correlated significantly with PD, but only when pictures portraying faculty alone and with their students are combined.

Table 12 Correlation Coefficients for research hypotheses

Cultural Dimension	Hypothesis	Coefficient	p-value
Power distance	H1: Symmetry correlated positively	.401	.325
	H2: Logos correlated positively	.765	.027
	H3: Images of faculty correlated positively, images of students correlated negatively	.237 .076	.573 .858
	H4: Images of buildings correlated positively, images of people correlated negatively	.364 -.249	.375 .552
Individualism/ Collectivism	H5: Images of individuals correlated positively, images of groups correlated negatively	.571 -.629	.186 .095
	H6: Images of students correlated positively, images of faculty correlated negatively	.266 -.327	.524 .429
	H7: Links correlated negatively	-.121	.776
Masculinity/ Femininity	H8: Figurative images correlated negatively, Buildings correlated positively	-.720 -.962	.044 .000
		.743	.035
	H9: High modality pictures correlated negatively Low modality pictures correlated positively	-.565 .369	.145 .369
	H10: Pictures of women correlated negatively	-.504	.203
	H11: Animated pictures correlated positively	.566	.144
Uncertainty Avoidance	H12: Links correlated negatively	-.450	.263
	H13: Horizontal pages correlated positively	-.110	.796
	H14: Abstract images correlated negatively	.255	.542

Although most of the correlations in Table 12 are too weak to be significant, all, with the exception of the UA measures, correlate in the hypothesized direction. This would suggest that Hofstede's dimensions are reflected in graphical design. They should not be treated as predictors of cultural preferences, however.

The results obtained for UA require further analysis. The fact that the directionality of correlation with horizontal pages was the opposite of what was hypothesized may be due to a mistaken assumption underlying the hypothesis itself. Countries with high UA index would prefer clear pages, with simple design, limited choices and restricted amount of data. It seemed that the horizontal pages—since they are smaller and do not require scrolling—would offer fewer choices. By using such aids as quick links and multiple layer menus, however, even horizontal pages can be made very interactive. Thus, number of links might be a better predictor of UA—that is, high UA countries would have fewer links (hypothesis 12).

Lack of abstract images was hypothesized to be another sign of high UA. But since abstract images were rare in the sample studied, it is difficult to draw a meaningful conclusion beyond the observation that the UA category is difficult to use for analysis of graphical representations.

The analysis also found another set of significant correlations that were not included in the hypotheses. These are summarized below:

- UA index correlated negatively with students ($r = -.710$, $p = .049$): countries with a high UA index had fewer numbers of pictures of students than countries with low UA;
- UA correlated negatively with photos ($r = -.781$, $p = .022$): countries with a high UA preferred drawings over photos;
- UA index correlated positively with buildings ($r = .870$, $p = .005$): countries with a high UA had more pictures depicting buildings;
- UA correlated negatively with persons ($r = -.949$, $p = .000$): pictures depicting people;
- IC correlated highly negatively with symmetry ($r = -.862$, $p = .006$): Individualistic countries (high IC) had fewer symmetrical pages than collectivist countries (low IC);
- IC correlated positively with images of females ($r = .764$, $p = .027$): Individualistic countries had more images of women.

This observation suggests that the list of culturally dependent graphical variables collected on the basis of available studies is incomplete and should be extended on the basis of Hofstede's original research and research conducted in other disciplines. Reference to other studies that use Hofstede's dimensions may shed some light on the interpretation of the additional correlations. For example, a *Reader's Digest* survey (1970, as cited in Hofstede, 2001) showed that high UA countries had significantly less favorable attitudes toward younger people ($r = -.77$) than countries with low UA. This supports the interpretation that the negative correlation of UA

and images of students in this study is not coincidental. It can also help us to understand the negative correlation between photos and UA that is otherwise rather surprising. Following Hofstede's research, one could assume that countries with a high UA index would prefer more realistic depictions and include more photographs on their pages. The results of the present research suggest the opposite. It may be in relation not to the medium itself (photo vs. drawing), but rather to what is represented, so that frequent photographs of students would not be uncommon in low UA countries.

Also, according to Hofstede's research, PD and UA were strongly positively correlated for European and Western cultures. Since the majority of the countries in the sample were Western countries, the above-mentioned strong correlations of images of buildings and people with UA may be an artifact of the countries chosen for the study. Similarly, the fact that PD and IC are negatively correlated (high PD countries tend to be collectivist), may partially explain the correlation between symmetry and the IC dimension. It was hypothesized that the pages in high PD countries would be more frequently symmetrical; this may explain why similar results are found for collectivist cultures.

There is no apparent explanation in the known literature for the finding that cultures with high individualism have university Web sites with more images of women.

Study Limitations

The results of this study are limited in several respects. The study researched only one genre, university Web sites. This design helped establish valid comparisons but also influences the results. For example, since the clients of universities are mostly young people regardless of country, it is natural that they will be a frequent subject of images.

Second, the number of countries was limited to eight, which can show similarities and differences among those cultures selected, but proved to be limited in testing whether cultural differences in Web sites can be measured/anticipated using Hofstede's dimensions of culture: Samples of 15 are recommended for calculating correlation coefficients. Including more countries and increasing the number of universities would help determine whether the results of this study are affected by the sample size.

Third, the study was limited to 20 sites per culture, a limitation difficult to overcome since some countries do not have more than 20 universities or universities with Web sites. It would be interesting to see the results of a study based on all the Web sites of the universities in each country.

A further limitation is the fact that the study analyzed only graphical elements. The literature suggests that cultural differences are also visible in language and discourse styles (Marcus & Gould, 2000; Rajkumar, 2003). Specific linguistic studies can be conducted only by people with a native or close to native knowledge of the

languages used. A collaborative cross-linguistic study might be attempted by a group of researchers from various countries.

Conclusions

The design of the university web sites studied here was found to vary across cultures. In each of the countries studied, certain characteristics occurred more frequently than in other countries: use of photos in Sweden, art depiction in Greece, use of animation in Malaysia and Ecuador, pastel colors in Japan, etc. The significant differences in number of links suggest different approaches to organizing and displaying information. Differences in colors schemes and image themes were also evident.

The Web sites also showed strong similarities across countries, as could be expected within a single genre. Most of the sites had a banner with the name of the university and its logo; students were most often the subject of the pictures (shown studying or laughing); and images of campus were common. Almost every institution treated the site as a visible representation of the university, its values, and commodities, promoting the image of a place where those values are fostered. Even without statistical analysis, however, one could notice that the values, and what follows, i.e., the promotional underlining, varied from country to country. This could be attributed to variations in education structure and in diversity of the types of the schools. For example, religious values were evident in sites from countries like Malaysia or Ecuador, where religious schools are common. In Sweden and Denmark, where performing arts are taught in special institutions, web sites with an artistic look were standard. Technical schools emphasized possibilities for professional advancement and school resources by including pictures of students working with high technology equipment. The older schools placed emphasis on tradition by using an official seal as a logo (one of the universities in Malaysia, Politeknik Seberag Perai, even included a page with an explanation of its significance) and by including university history. In some cases, the university seal covered a large portion of the page. The newer institutions had logos with modern designs; their sites placed weight on future opportunities for their students.

The observed similarities and differences in web design can be seen in terms of the dimensions of Hofstede's cultural model. The cultural markers identified in the literature seem to relate to specific dimensions, although the list of relevant cultural markers is not yet complete. The correlations were much weaker than anticipated, however. The weak correlations can be interpreted in several ways. The first is that Hofstede's model may not be valid, as some of its critics suggest. The second possibility is that Hofstede's model may be valid but that the cultural markers associated with each or some of the dimensions are based on faulty assumptions, and that Hofstede's model is not applicable to the study of graphical representations. Third, it is possible that Hofstede's model is valid, and can be used in studying differences in web design, but that the limitations of the present study—including the small

number of countries studied—do not show the full potential of the model. In as much as most correlations did occur in the hypothesized direction, it would seem that the last interpretation is most likely, and that Hofstede's model may be helpful in interpreting cultural similarities and differences in graphical design. However, Hofstede's dimensions and graphical cultural markers require more intensive study, especially when interpreting UA. Therefore, Hofstede's model should be used with caution by researchers and/or developers attempting to construct culturally appropriate web sites.

It is also important to bear in mind that culture may not be the only source of variation. Observational study involving only a survey of graphical elements does not allow us to determine who the designers are, what motivated their choices, etc. In particular we need to ask whether they were influenced by the design of other university Web sites or required to follow institutional guidelines in promoting their university's image. It is also possible that the availability of online tutorials and online Webmasters' discussion groups may have influenced the ideas represented in the design. Economic factors could also play a role; poorer universities may not be able to acquire more sophisticated Web development software or have a highly trained Web design staff. This last possibility did not seem to be true for the sites studied here, since Hispanic sites were heavily animated with pictures and text, while Scandinavian sites strove for simplicity. Also, the Ecuadorian sites ranged in complexity from simple to sophisticated, although some of them had many glitches, suggesting that the designers were not professionals. For their part, Japanese sites seem to be designed on much larger monitors, since most of them required horizontal scrolling when viewed on 13-inch and some even with 16-inch monitors.

But the most important questions about culture and web design must be asked from the perspective of the user: Do the cultural differences evident in graphical design influence Web site usability? Do the people from the countries from opposite poles of Hofstede's dimensions prefer Web sites with widely divergent designs, and more importantly, do they perform better on the interfaces designed specifically for their culture? These questions point to the need for further studies to examine users from other cultures, their expectations and needs, and their aesthetic preferences.

Notes

- 1 ANOVA (ANALYSIS OF VARIANCE) is a calculation procedure that compares means of two or more groups in order to determine whether the differences among the means are statistically significant or could occur by chance. It compares the how far apart the means of all groups are with how much variation there is within each group. A large value of F indicates that there is greater difference between groups than within groups. The lower the value of p , the lower the probability that the received results occurred by chance. Since an ANOVA test will answer only whether the differences among means exist or not, other procedures are needed to determine where the differences occurred and which groups are particularly different from each other. The Bonferroni test is one such procedure.

- 2 [Special issue editor's note: Scandinavians are not likely to be surprised by these findings. Similarities between Japanese and Swedish cultures—in clear contrast with Danish and Norwegian cultures—have been documented (Burton, 1989). More broadly, in my own experience, each cultural group is acutely aware of the distinctions that set them apart from their Scandinavian kin—and is eager to maintain those distinctions. These begin with language: While Swedes and Norwegians can largely understand one another, despite strong differences between the two, they both complain about the impossibility of Danish. Danes generally seem to have little trouble with either Swedish or Norwegian. While any characterizations in a short space will run the risk of stereotyping and essentializing, it may be safe to say that that Danes, at least in the urban areas, regard themselves as the most cosmopolitan of the Scandinavians (reflected, for example, in Denmark being the first country to legalize pornography—for which it is still [in]famous elsewhere in Scandinavia). Danes tend to see Swedes as correspondingly more formal, and Norwegians as more darkly pious and conservative—even though all countries share Lutheranism as the state church. See Rasmussen & Bank-Middelsen, 2000, for an initial discussion.]

References

- Ackerman, S. K. (2002). *Mapping user interface design to culture dimensions*. Paper presented at International Workshop on Internationalization of Products and Systems, Austin TX, USA. Retrieved May 20, 2002, from http://www.iwips2002.org/downloads/AMA_XCult_13Jul02.ppt
- Albers-Miller, N. D., & Gelb, B. D. (1996). Business advertising appeals as a mirror of cultural dimensions: A study of eleven countries. *Journal of Advertising*, 25, 57–70.
- Al-Olayan, F. S., & Karande, K. (2000). A content analysis of magazine advertisements from the United States and the Arab world. *Journal of Advertising*, 29, 69–83.
- Barber, W., & Badre, A. (1998). *Culturability: The merging of culture and usability*. Paper presented at 4th Conference on Human Factors & the Web, Baskin Ridge NJ, USA. Retrieved April 30, 2000, from <http://www.research.att.com/conf/hfweb/proceedings/barber/>
- Baskerville, R. F. (2003). Hofstede never studied culture. *Accounting, Organizations and Society*, 28, 1–14.
- Bell, P. (2001). Content analysis of visual images. In T. van Leeuwen & C. Jewitt (Eds.), *Handbook of visual analysis* (pp. 92–118). London: Sage.
- Burton, J. [1989] 1991). Europe's blond, blue-eyed JAPANESE. *International Management* (May). Reprinted in B. Arildsen, O. Faaborg & R. Flensted (Eds.), *Across cultures: Cultural management* (pp. 77–80). Herning, Denmark: systime.
- Bucher, H.-J. (2002). The power of the audience: Interculturality, interactivity and trust in Internet communication. In F. Sudweeks & C. Ess (Eds.), *Proceedings cultural attitudes towards communication and technology 2002* (pp. 3–14). Murdoch, Western Australia: School of Information Technology, Murdoch University.
- Callahan, E. (2004). Interface design and culture. *Annual Review of Information Science and Technology*, 39, 257–310.
- Choong, Y. Y., & Salvendy, G. (1999). Implications for design of computer interfaces for Chinese users in mainland China. *International Journal of Human-Computer Interaction*, 11, 29–46.

- Cutler, B. D., & Javalgi, R. (1992). A cross-cultural analysis of the visual components of print advertising: The United States and the European Community. *Journal of Advertising Research*, 32, 71–80.
- del Galdo, E. (1990). Internationalization and translation. Some guidelines for the design of human-computer interfaces. In J. Nielsen (Ed.), *Designing User Interfaces for International Use* (pp. 1–10). Amsterdam: Elsevier.
- Dormann, C., & Chisalita, C. (2002). *Cultural values in web site design*. Paper presented at ECCE-11: Eleventh European Conference on Cognitive Ergonomics, September 8–11, 2002, Catania, Italy. Retrieved Feb.10, 2003, from <http://www.cs.vu.nl/~claire/Hofstede-dormann.pdf>
- Duncker, E., Theng, Y-L., & Mohd-Nasir, N. (2000). Cultural usability in digital libraries. *ASIS Bulletin* 26 (4). Retrieved April 14, 2003, from http://www.asis.org/Bulletin/May-00/duncker_et_al.html
- Evans, M., McBride, A., Queen, M., Thayer, A., & Spyridakis, J. (2004). Has the tone of online English become globalized? An empirical research study investigating the written tone of university web sites around the world. In F. Sudweeks and C. Ess (Eds.), *Fourth International Conference on Cultural Attitudes towards Technology and Communication 2004* (pp. 135–139). Murdoch, Western Australia: School of Information Technology, Murdoch University.
- Fang, T. (2003). Critique of Hofstede's fifth national culture dimension. *International Journal of Cross Cultural Management* 3, 347–368.
- Hofstede, G. (1980). *Culture's consequences: International differences in work-related values*. Newbury Park, CA: Sage.
- Hofstede, G. (1991). *Cultures and organizations: Software of the mind*. London: McGraw-Hill.
- Hofstede, G. (2001). *Culture's consequences: International differences in work-related values*. Newbury Park, CA: Sage.
- Hofstede, G. (2002). Dimensions do not exist: A reply to Brendan McSweeney, *Human Relations*, 55, 1355–1361.
- Leonardi, P. (2002). Cultural variability in web interface design: Communicating US Hispanic cultural values on the Internet. In F. Sudweeks & C. Ess (Eds.), *Proceedings Cultural Attitudes Towards Communication and Technology 2002* (pp. 297–315). Murdoch, Western Australia: School of Information Technology, Murdoch University.
- Lin, S.-L. (2003). *Websites design of Internet banks in U.S., U.K. and Japan*. Unpublished manuscript, School of Library and Information Science, Indiana University Bloomington.
- Marcus, A. & Gould, E. W. (2000, June 19). Cultural dimensions and global Web user-interface design: What? So what? Now what? In *Proceedings of the 6th Conference on Human Factors and the Web, June 19, 2000*. Retrieved Jun. 15, 2001, from <http://www.amanda.com/resources/hfweb2000/hfweb00.marcus.html>
- McSweeney, B. (2002a). Hofstede's model of national cultural differences and consequences: A triumph of faith - a failure of analysis. *Human Relations*, 55, 89–118.
- McSweeney, B. (2002b). The essentials of scholarship: A reply to Geert Hofstede. *Human Relations*, 55, 1363–1372.
- Rajkumar, S. (2003). *University Web sites: Design differences and reflections of culture*. Unpublished manuscript, School of Library and Information Science, Indiana University Bloomington.

- Rasmussen, A. S., & Bank-Middelsen, G. (2000). *Focus on culture*. Århus, Denmark: systime.
- Reader's Digest. (1970). *A Survey of Europe Today*. London: Author.
- Robbins, S. S., & Stylianou, A. C. (2002). A study of cultural differences in global corporate web sites. *Journal of Computer Information Systems*, 42, 3–9.
- Roberts, K. H., & Boyacigiller, N. (1984). Cross national organizational research: The grasp of the blind men. In L. L. Cummings & B. M. Staw (Eds.), *Research in organizational behavior*, vol. 6 (pp. 455–488). Greenwich, CT: JAI Press.
- Russo, P., & Boor, S. (1993). *How fluent is your interface? Designing for international users. Proceedings INTERCHI '93 Conference on Human Factors in Computing Systems: INTERACT '93 and CHI '93. Amsterdam, 24–29 April* (pp. 342–347). ACM Press.
- Schmid-Isler, S. (2000). The language of digital genres—a semiotic investigation of style and iconology on the World Wide Web. *Proceedings of the 33rd Hawaii International Conference on System Sciences (HICSS 2000)*. Retrieved October 6, 2005, from <http://csdl2.computer.org/comp/proceedings/hicss/2000/0493/03/04933012.pdf>
- Sheppard, C., & Scholtz, J. (1999). The effects of cultural markers on Web site use. In *Proceedings of the 5th Conference on Human Factors & the Web, Gaithersburg, Maryland, USA*. Retrieved September 15, 2001, from <http://zing.ncsl.nist.gov/hfweb/proceedings/sheppard/>
- Smith, P. B. (2002). Culture's consequences: Something old and something new. *Human Relations*, 55, 119–135.
- Søndergaard, M. (1994). Hofstede's Consequences: A study of reviews, citations and replications. *Organisation Studies*, 15, 447–456.
- Tansey, R., & Hyman, M. R. (1990). Cultural themes in Brazilian and U.S. auto ads: A cross-cultural comparison. *Journal of Advertising*, 19, 30–40.
- Tractinsky, N. (1997). Aesthetics and apparent usability: Empirically assessing cultural and methodological issues. In *Conference proceedings on Human Factors in Computer Systems* (pp. 115–122). New York, NY: ACM Press.
- Wernick, A. (1991). *Promotional culture: Advertising, ideology and symbolic expression*. London: Sage.
- Williamson, D. (2002). Forward from a critique of Hofstede's model of national culture. *Human Relations*, 55, 1373–1395.

About the Author

Ewa Callahan is a Ph.D. candidate in information science at Indiana University. Her dissertation examines similarities and differences in web design among world cultures. Her general research and teaching interests are in the areas of international issues in information science, human computer-interaction, information visualization, and development and use of information technologies in postcommunist countries.

Address: 864 Flatbush Ave., West Hartford, CT 06110 USA